

Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R i_1(s) + L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)]$$

$$L s [i_1(s) - i_2(s)] + R [i_1(s) - i_2(s)] = R i_2(s) + R i_2(s) + \frac{i_2(s)}{C s}$$

$$V_s(s) = R i_2(s) + \frac{i_2(s)}{C s} = \left(\frac{R C s + 1}{C s} \right) i_2(s)$$

Procedimiento algebraico

$$V_e(s) = (L s + 2 R) i_1(s) = (L s + R) i_2(s)$$

$$(L s + R) i_1(s) = \left(\frac{C L s^2 + 3 C R s + 1}{C s} \right) i_2(s)$$

$$i_1(s) = \left(\frac{C L s^2 + 3 C R s + 1}{C s} \right) \frac{1}{L s + R} i_2(s) = \left(\frac{C L s^2 + 3 C R s + 1}{C s (L s + R)} \right) i_2(s)$$

$$V_e(s) = (L s + 2 R) \left(\frac{C L s^2 + 3 C R s + 1}{C s (L s + R)} \right) i_2(s) - (L s + R) i_2(s)$$

$$V_e(s) = \frac{[L s + 2 R][C L s^2 + 3 C R s + 1] - C s [L s + R]^2}{C s [L s + R]} i_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{\left(\frac{R C s + 1}{C s} \right) i_2(s)}{\frac{[L s + 2 R][C L s^2 + 3 C R s + 1] - C s [L s + R]^2}{C s [L s + R]} i_2(s)} = \frac{[L s + R][R C s + 1]}{[L s + 2 R][C L s^2 + 3 C R s + 1] - C s [L s + R]^2}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{L R C s^2 + (R^2 C + L) s + R}{C L^2 s^3 + 2 R C L s^2 + 3 R C L s^2 + 6 R^2 C s + L s + 2 R - C L^2 s^3 - 2 R C L s^2 - R^2 C s}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{RCLs^2 + (R^2C + L)s + R}{3RCLs^2 + (SR^2C + L)s + 2R}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} \frac{1}{s} \left(1 - \frac{V_s(s)}{V_e(s)} \right) = \lim_{s \rightarrow 0} \frac{1}{s} \left(1 - \frac{RCLs^2 + (R^2C + L)s + R}{3RCLs^2 + (SR^2C + L)s + 2R} \right) = 1 - \frac{1}{2} = 0.5V$$

Estabilidad en lazo abierto

$$R = 2.2k$$

$$a = 3(2.2 \times 10^3)(470 \times 10^{-6})(22 \times 10^{-6})$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$L = 22m$$

$$b = [5(2.2 \times 10^3)^2(470 \times 10^{-6})] + 22 \times 10^{-3}$$

$$C = 470nF$$

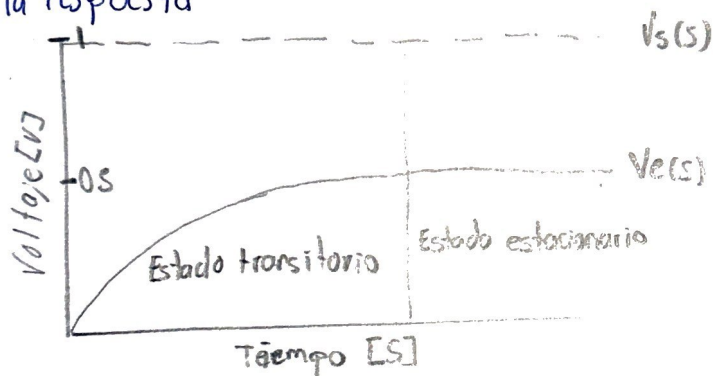
$$c = 2(2.2 \times 10^3)$$

$$\lambda_1 = -0.386$$

$$\lambda_2 = -166.66 \times 10^6$$

Sistema estable con una respuesta sobreamortiguada

Bosquejo de la respuesta



Modelo de ecuaciones integro-diferenciales

$$V_e(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[V_e(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{di_1(t) - i_2(t)}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_s(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$